MCTA 3203

MECHATRONICS SYSTEM INTEGRATION

SEMESTER 1, 25/26

WEEK 07 : PLC INTERFACING WITH MICROCONTROLLER AND PC OVER ETHERNET/IP

SECTION 2

GROUP 19

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ABSTRACT

This experiment investigates the integration of a Programmable Logic Controller (PLC) with an Arduino microcontroller using the OpenPLC platform. The objective was to design, simulate, and implement ladder logic programs capable of controlling hardware components through digital input and output mappings. The practical tasks involved a Start–Stop control circuit using pushbuttons and an LED. The ladder diagrams were developed and tested in OpenPLC Editor before being uploaded to the Arduino board for real-time execution. The results confirm that the microcontroller responded accurately to the PLC logic, with successful latching behavior observed in the Start–Stop circuit. The experiment demonstrates essential concepts in industrial automation, including PLC programming, hardware interfacing, and digital control logic.

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1.0 INTRODUCTION

Programmable Logic Controllers (PLCs) are widely used in industry to control machines and automated systems. In this lab, we explored how a PLC programmed using the OpenPLC Editor that can be connected to both a microcontroller and a computer over Ethernet/IP. By using OpenPLC together with an Arduino board, we were able to design a simple ladder logic program, simulate it, and then upload it to real hardware.

The lab began with creating a basic blinking LED circuit. This gave us an introduction to the OpenPLC environment, including how to build ladder diagrams, assign variables, and test the logic through simulation. After that, we moved on to a more practical example: building a Start–Stop control circuit. This exercise helped us understand how PLC logic interacts with physical components such as push buttons and LEDs. Overall, the experiment gave us hands-on experience in PLC programming and hardware interfacing, which is essential in mechatronics and automation engineering.

2.0 MATERIALS AND EQUIPMENT

Item	**Description**
Laptop	Running OpenPLC Editor
Arduino Board	Arduino acting as PLC runtime
Pushbutton (Start)	NO type, connected to Digital Pin 3
Pushbutton (Stop)	NO type (actual wiring), connected to Digital Pin 4
Pull-down Resistors	Connected to GND for both buttons
LED	Connected to Digital Pin 7 via 220 Ω resistor

220 Ω Resistor	LED current limiting resistor
Breadboard	Circuit mounting
Jumper wires	Signal connections

3.0 EXPERIMENTAL SETUP

1. OpenPLC Editor was installed and a new ladder project was created.

2. Variables were added:

UP_Start → %IX0.0 (mapped to Arduino D3)

DOWN_Stop → %IX0.1 (mapped to Arduino D4)

Latch → Internal memory bit

LED → %QX0.0 (mapped to Arduino D7)

3. Breadboard wiring:Both pushbuttons were configured using pull-down resistors to ground, making them NO inputs. LED was wired to pin 7 with a 220 Ω resistor.

4. Arduino was connected to the laptop and set as the OpenPLC runtime.

5. The ladder diagram was simulated before being uploaded to hardware.

4.0 METHODOLOGY

Step 1 — Create Variables

Variables were defined in OpenPLC Editor and assigned to physical pins:

%IX0.0 → D3 (Start)

%IX0.1 → D4 (Stop)

%QX0.0 → D7 (LED)

Step 2 — Construct Ladder Logic

The logic uses:

Start (NO contact)

Stop (NC contact) in theory

But in hardware, since Stop is wired as NO, the logic functions based on HIGH when pressed.

Latch contact in parallel with Start to maintain output

Output coil controlling the LED

Step 3 — Upload to Arduino

After compiling, the program was transferred to Arduino through COM port.

Step 4 — Hardware Testing

Four tests were performed:

1. Press Start → LED turns ON
2. Release Start → LED remains ON
3. Press Stop → LED turns OFF
4. Test simultaneous press
5. Observe software debug and hardware output behavior

5.0 DATA COLLECTION

Trial	**Time**	**Start (Pin 3)**	**Stop (Pin 4)**	**Latch State**	**LED (Pin 7)**	**Notes**
1	14:40:03	LOW (Released)	LOW (Released)	False	Off	Initial state
2	14:40:12	HIGH (Pressed)	LOW (Released)	True	On	Start pressed
3	14:40:20	LOW (Released)	LOW (Released)	True	On	Start released, LED latched
4	14:40:30	LOW (Released)	HIGH (Pressed)	False	Off	Stop pressed
5	14:40:45	HIGH (Pressed)	HIGH (Pressed)	False	Off	Both pressed

6.0 DATA ANALYSIS

Start Button Behavior

Since Start is NO with pull-down:

Press → input becomes HIGH

LED latches ON

Stop Button Actual Behavior

Although industrial circuits use NC Stop, this setup used a NO Stop with pull-down.
This means:

Press Stop → input becomes HIGH

Logic interpreted this press as a “Stop” signal

LED turns OFF

Latch Operation

The latch contact successfully maintained the LED ON after Start was released, confirming correct self-hold logic.

Simultaneous Press

Stop condition dominated, turning off the latch—expected behavior.

7.0 RESULT

The outcome of the experiment shows that programmed ladder diagrams operated successfully once uploaded to the Arduino board.

Pressing the START button activated the latch and kept the LED ON continuously, while pressing the STOP button broke the latch and turned the LED OFF. The LED responded immediately to each button input, confirming proper variable assignment, wiring, and COM port configuration.

Overall, the system performed as expected, with smooth operation and accurate execution of the ladder logic. Minor issues only occurred when incorrect pins were assigned, but these were resolved during troubleshooting.

8.0 DISCUSSION

Throughout the experiment, we followed the complete flow of designing, testing, and implementing a PLC program. Creating the ladder diagram for the blinking LED helped us learn how to use contacts, coils, and function blocks in OpenPLC. The simulation step was especially helpful because it allowed us to check if the program worked properly before uploading it to the Arduino.

A key part of the experiment was mapping the OpenPLC variables to the correct Arduino pins. Without proper pin assignment, the microcontroller would not respond to the signals from the ladder logic. We also had to ensure that the correct COM port was selected before transferring the program. Once everything was set up, the Arduino successfully ran the PLC logic and controlled the LED as expected.

The Start–Stop control circuit gave us deeper insight into how PLCs are used in real industrial systems. The latching mechanism in the ladder diagram allowed the LED to stay ON after pressing the START button, and turn OFF using the STOP button. Building and wiring the circuit on the breadboard showed how software logic and hardware must work together correctly. During testing, we noticed that even small wiring mistakes or wrong variable assignments could cause the system to malfunction, highlighting the importance of accuracy when working with control systems.

9.0 CONCLUSION

This experiment successfully demonstrated how PLC ladder logic can be programmed, simulated, and executed on a microcontroller using the OpenPLC platform. The Start–Stop control circuit functioned correctly, proving that the Arduino was able to interpret and run PLC logic in real time. The experiment reinforced key concepts in PLC design, including contact–coil operations, variable mapping, and digital control. It also highlighted the importance of accurate wiring and correct COM port selection when interfacing PLC software with hardware. Overall, the activity achieved its objectives and provided valuable foundational experience in industrial automation and mechatronic system integration.

10.0 RECOMMENDATIONS

For future activities, it would be helpful to check all wiring and pin assignments carefully to avoid problems during testing. Students can also explore more PLC features—such as timers and counters—to build more advanced and realistic control circuits. Learning how to use Ethernet/IP communication in the next lessons would also be useful, as it reflects how modern industrial devices connect and share data today. Adding simple safety elements, like an emergency stop button, could make the experiments even more practical and similar to real industrial applications.

11.0 REFERENCES

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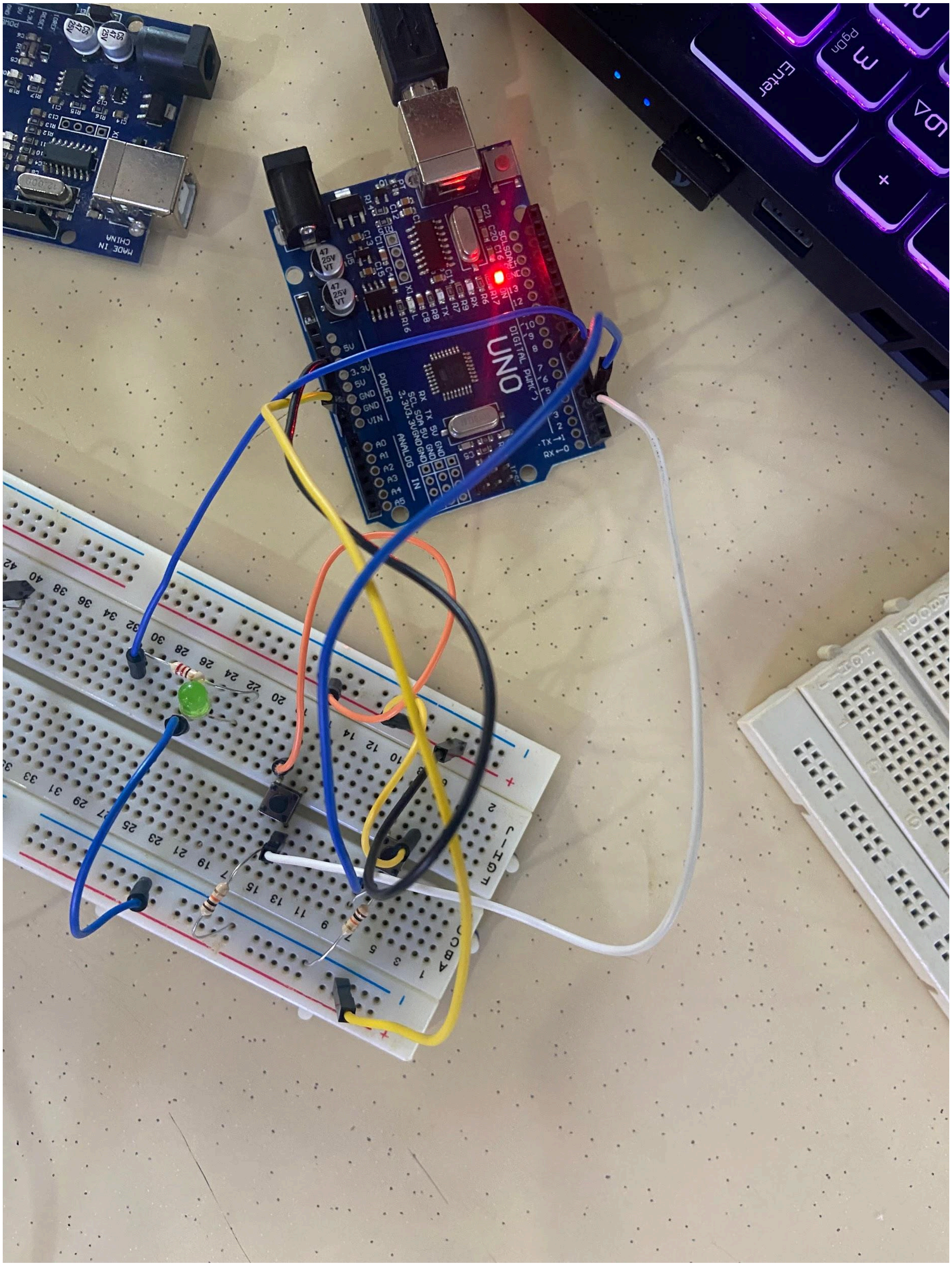
APPENDICES



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STUDENTS'S DECLARATION

<u>CERTIFICATE OF ORIGINALITY AND AUTHENTICITY</u>

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specific in references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or person.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report.** The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of individual's contributor to the report.

We, therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** have been **verified by us.**

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